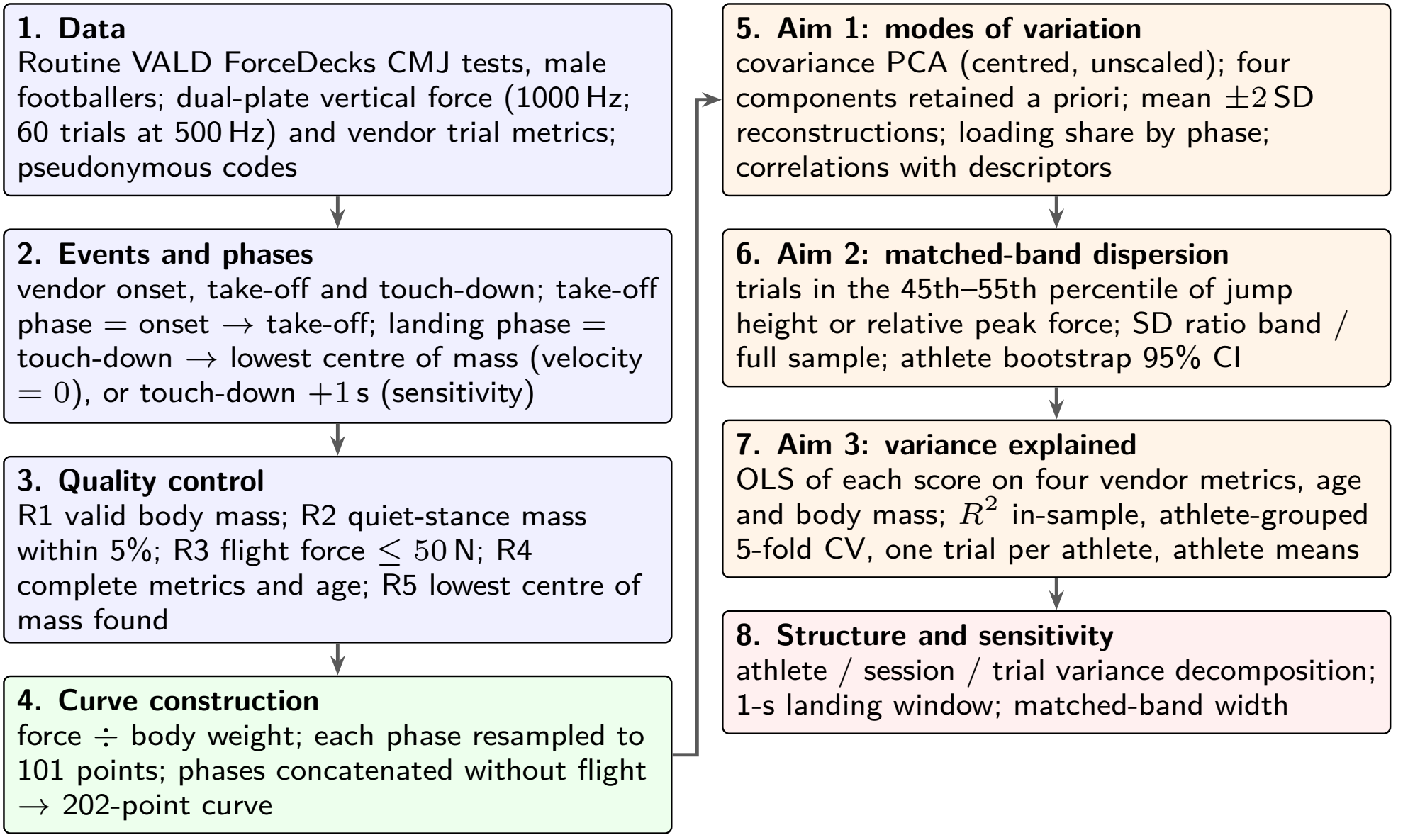
1. Data
Routine VALD ForceDecks CMJ tests, male footballers; dual-plate vertical force (1000 Hz; 60 trials at 500 Hz) and vendor trial metrics; pseudonymous codes
2. Events and phases
vendor onset, take-off and touch-down; take-off phase = onset → take-off; landing phase = touch-down → lowest centre of mass (velocity = 0), or touch-down +1 s (sensitivity)
3. Quality control
R1 valid body mass; R2 quiet-stance mass within 5%; R3 flight force ≤ 50 N; R4 complete metrics and age; R5 lowest centre of mass found
4. Curve construction
force ÷ body weight; each phase resampled to 101 points; phases concatenated without flight → 202-point curve
5. Aim 1: modes of variation
covariance PCA (centred, unscaled); four components retained a priori; mean ±2 SD reconstructions; loading share by phase; correlations with descriptors
6. Aim 2: matched-band dispersion
trials in the 45th–55th percentile of jump height or relative peak force; SD ratio band / full sample; athlete bootstrap 95% CI
7. Aim 3: variance explained
OLS of each score on four vendor metrics, age and body mass; R^2 in-sample, athlete-grouped 5-fold CV, one trial per athlete, athlete means
8. Structure and sensitivity
athlete / session / trial variance decomposition; 1-s landing window; matched-band width